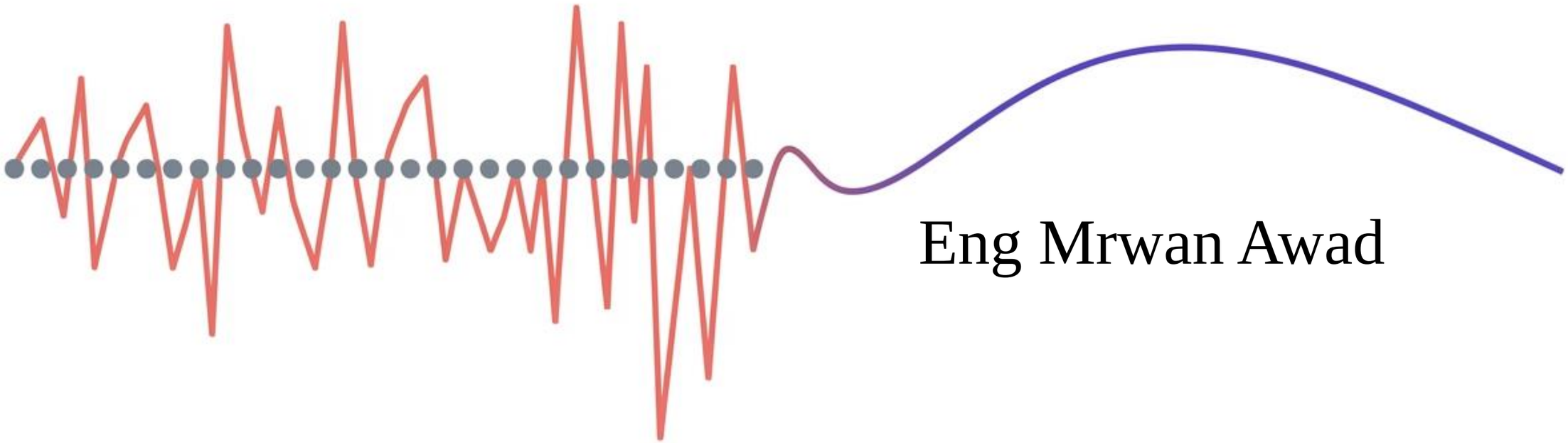
Mastering Regularization: L1 & L2

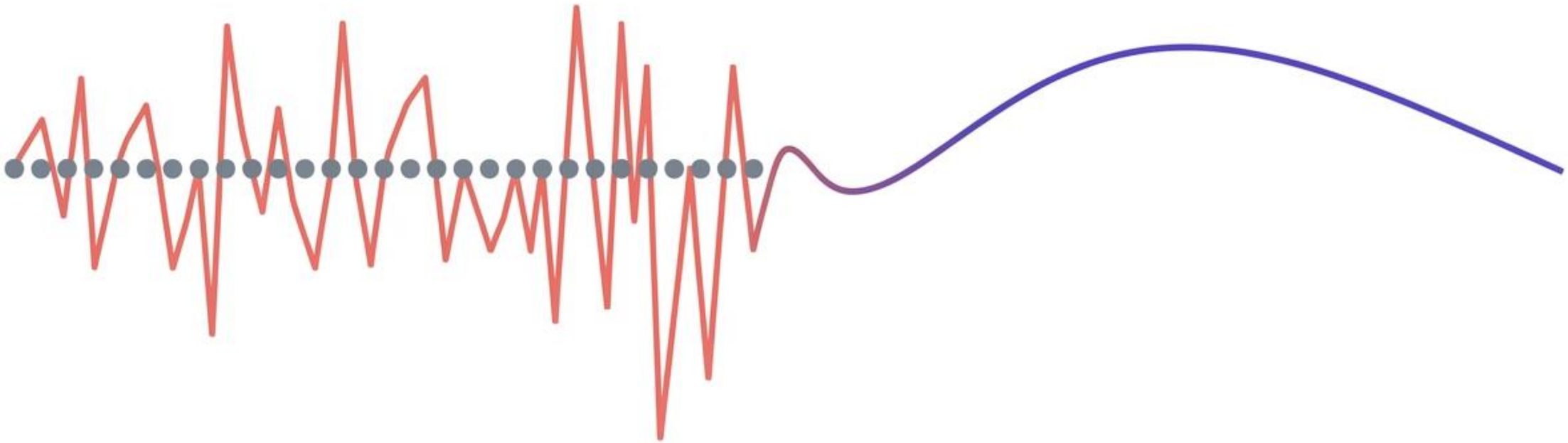
Diagnosing overfiting and implemeting mathematical penalties (Lasso & Ridge) to improve model generalization.



Eng Mrwan Awad

Overfitting occurs when a model memorizes noise.

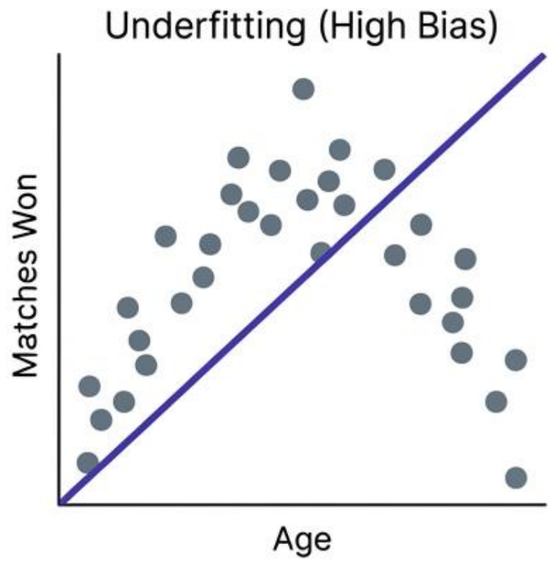
Bias vs Variance Tradoff



What does <u>high bias</u> means ?

Bias measures:

How far the average prediction of the model is from the true function.



<u>Ex:</u>

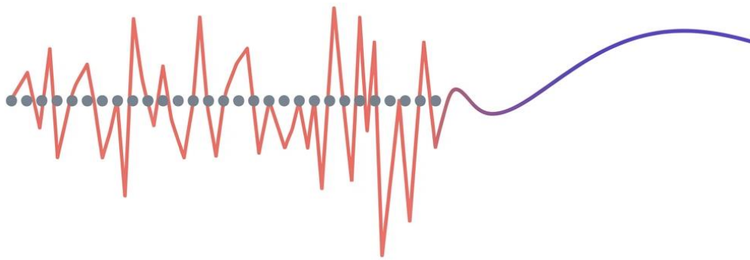
True relationship is quadratic: $y = x^2$

But the *<u>model fit</u>*:

y=b0+b1x

<u>How to know ?</u>

- Training error is high
- Test error is also high
- Model too simple



For Example

Can be calculated by :

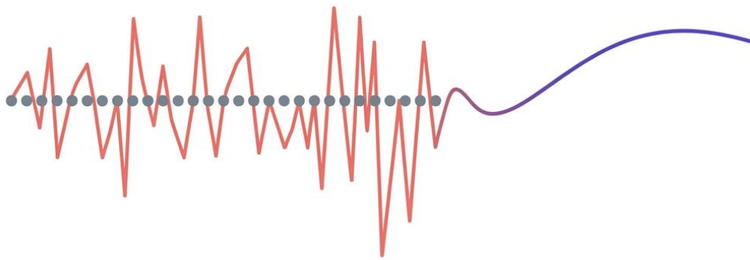
$$\boxed{Y = 2x}$$

Suppose this is our problem :

X	Y (True)
1	2
2	4
3	6

Suppose our model predicts :

$$\hat{Y} = x$$



For Example

After calculating y_pred :

X	Y (True)	$\hat{Y}$ (pred)
1	2	1
2	4	2
3	6	3

Avg. prediction = $\frac{1+2+3}{3}$

$\boxed{3}$

Is 3 far from true values ?

Yes

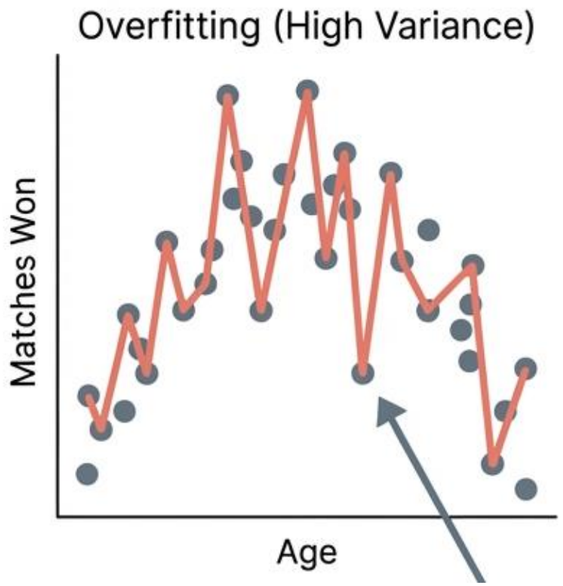
<u>High Bias</u>

What does high Variance means ?

Variance measures:

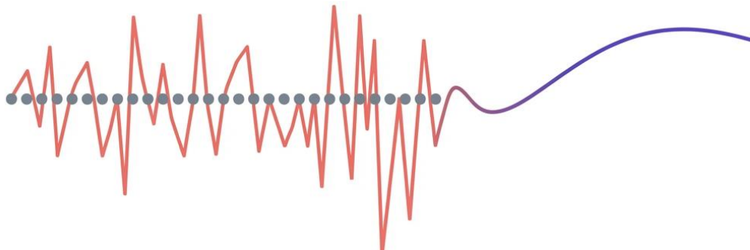
How much the model changes if we train it on different datasets.

How far predictions move around their average ?

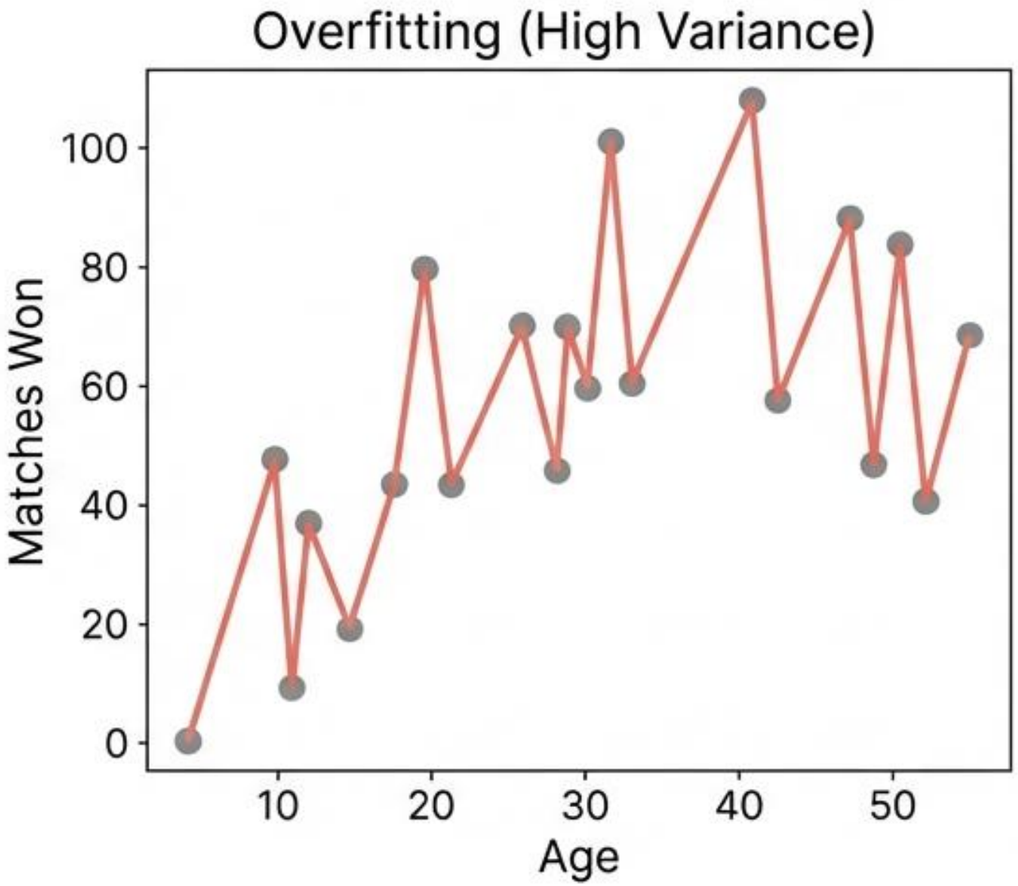


How to know ?

- Training error is low
- Test error is also high
- Model captures noise (doesn't generalize)



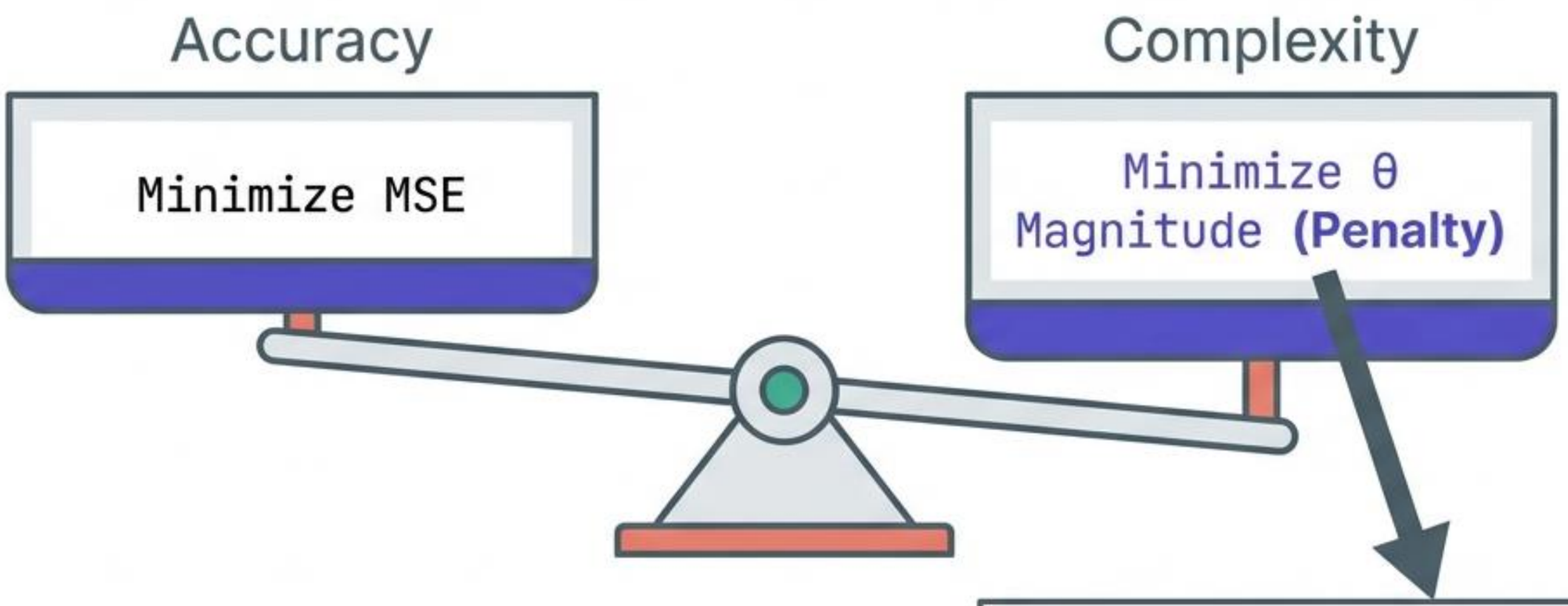
Complex boundaries are driven by high-magnitude coefficients.



$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4 \ldots$$

The optimization algorithm inflates parameters (θ) to twist the line through every training point. To fix this, we must suppress these values.

Regularization adds a penalty cost to model complexity.



The **Penalty Term** forces the model to keep coefficients small.

Small coefficients = Simpler models = Better generalization

Transforming the Cost Function

$$J(\theta) = MSE + \lambda \times (\text{Penalty Term})$$

Standard Error calculation (Residuals).

The Tuning Knob
High λ = High Penalty (Risk of Underfitting).
Low λ = Low Penalty (Risk of Overfitting).

Deep Dive: L2 Regularization (Ridge)

Minimizes the SQUARE of the coefficients.

$$\text{Penalty} = \lambda \sum (\theta^2)$$

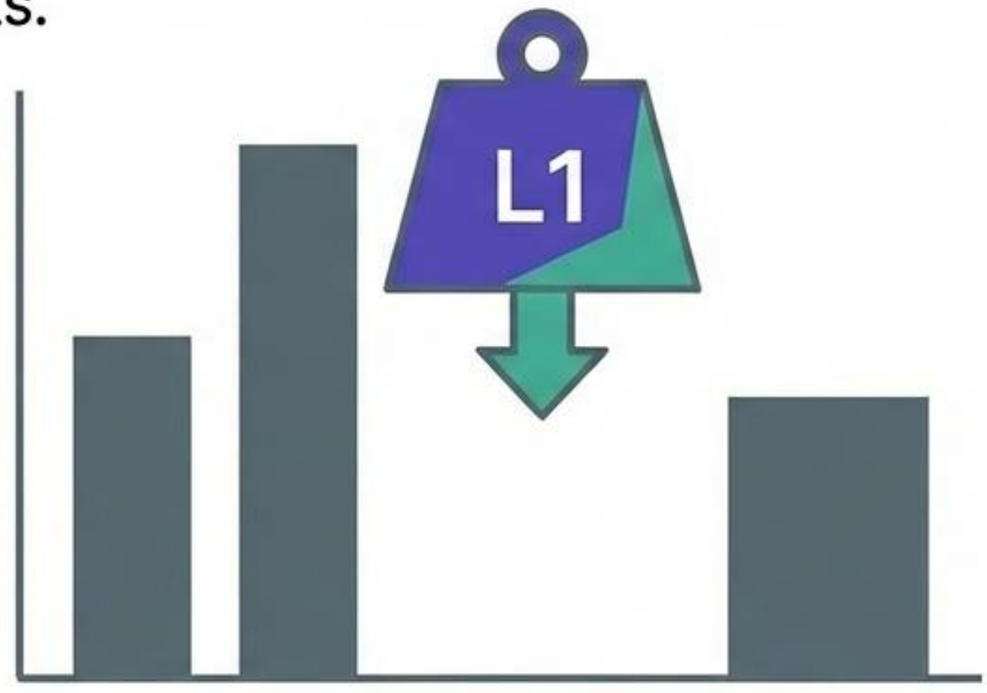
Because the term is squared, outliers have a huge impact. Ridge shrinks coefficients aggressively toward zero, but rarely exactly TO zero.

`sklearn.linear_model.Ridge`

Deep Dive: L1 Regularization (Lasso)

Minimizes the ABSOLUTE value of the coefficients.

$$\text{Penalty} = \lambda \sum |\theta|$$



L1 can drive coefficients to exactly zero. This effectively performs Feature Selection by removing less important variables entirely.

`sklearn.linear_model.Lasso`

Case Study: Predicting Melbourne Housing Prices

Suburb	Rooms	Type	Method	Postcode	Regionname	Propertycount	Distance	CouncilArea	Price
Abbotsford	2	h	SS	3067	Northern Metropolitan	4019	2.5	Yarra	1480000
Richmond	3	h	S	3121	Northern Metropolitan	4019	3.1	Yarra	1100000
Collingwood	2	u	SP	3066	Northern Metropolitan	4019	2.0	Yarra	650000
Fitzroy	3	h	PI	3065	Northern Metropolitan	4019	1.8	Yarra	1250000
Cardenond	3	h	S	3067	Northern Metropolitan	4019	3.1	Yarra	1100000
Tayword	4	u	SP	3121	Northern Metropolitan	4019	2.0	Yarra	950000
Abbotsford	2	h	PI	3065	Northern Metropolitan	4019	1.8	Yarra	1600000

Source:
Kaggle

Records:
34,857

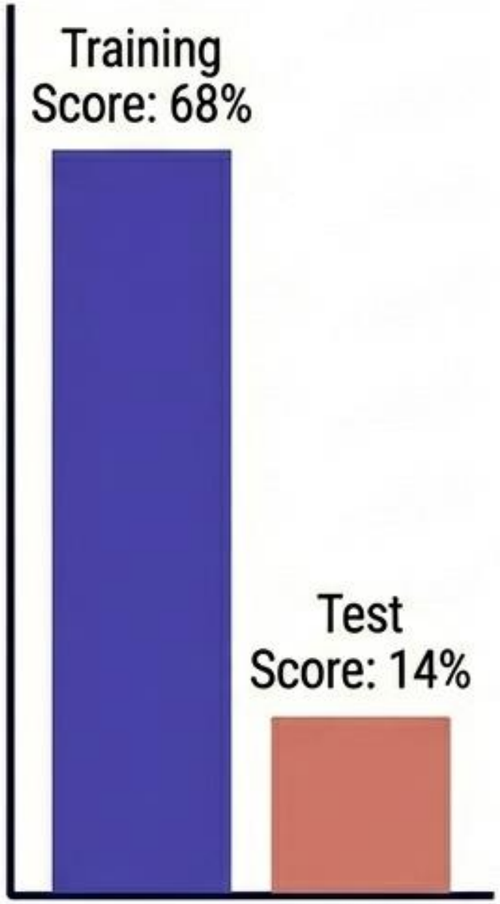
Initial Columns:
21

Target Variable:
Price

Linear Regression.

```
reg = LinearRegression().fit(X_train, y_train)

reg.score(X_test, y_test)
```



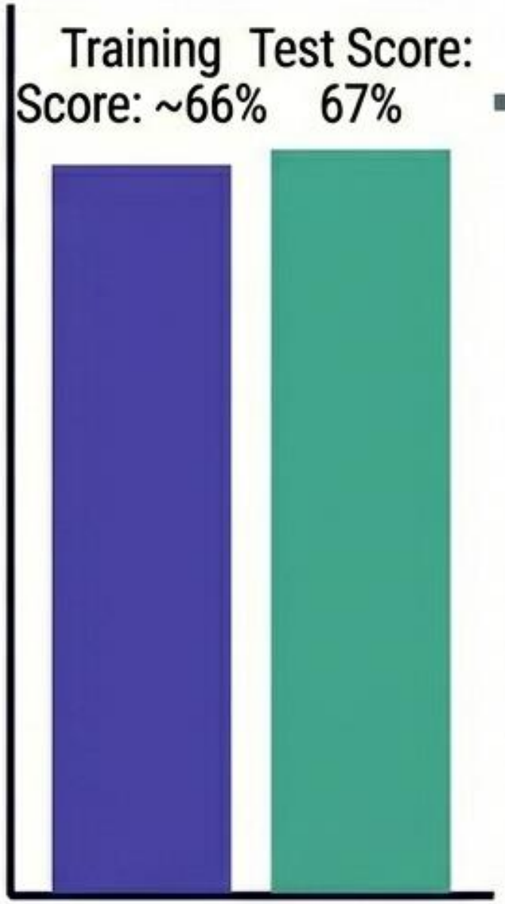
SEVERE OVERFITTING.
The model failed to generalize.

Solution 1: Applying L1 (Lasso).

```
from sklearn.linear_model import Lasso

lasso_reg = Lasso(alpha=50, max_iter=100, tol=0.1)

lasso_reg.fit(X_train, y_train)
```



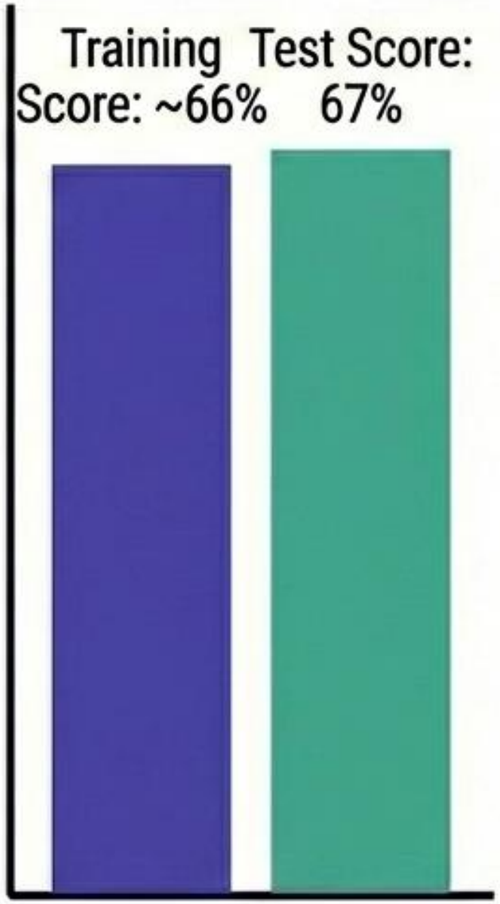
Massive improvement in generalization. Lasso ignored the noise in the **sparse dummy variables**.

Solution 2: Applying L2 (Ridge).

```
from sklearn.linear_model import Ridge

ridge_reg = Ridge(alpha=50, max_iter=100, tol=0.1)

ridge_reg.fit(X_train, y_train)
```



L2 yields
L2 yields similar results to L1 here. **squared penalty** effectively curbed the high coefficients caused by the **dimensional explosion**.

Comparative Analysis

Model	Penalty Math	Test Score	Outcome
Linear Regression	None	14%	Severe Overfitting
Lasso (L1)	Absolute Value $\|\theta\|$	67%	Feature Selection
Ridge (L2)	Squared Value θ^2	67%	Coefficient Shrinkage

NotebookLM